

Scale SEAL Leaderboards

Frontier Model Performance Across 20+ Benchmarks

Audience: ML Researchers, AI Lab Leadership, Engineering Managers

Core Thesis: No single lab wins everywhere; specialization is defining frontier competition.

OpenAI

Google

Anthropic

Muse Spark

Benchmark Breadth: Four Capability Domains

Scale evaluates 100+ models across private + public datasets with confidence intervals.

Agentic

Coding, repo Q&A, tool use

SWE Atlas, SWE-Bench Pro, MCP Atlas

Frontier

Reasoning & knowledge limits

HLE, EnigmaEval, MultiChallenge, SciPredict

Safety

Honesty, misuse propensity, professional reasoning

MASK, Fortress, PropensityBench, Legal/Finance

Multimodal

Audio + vision-language understanding

AudioMultiChallenge, VISTA, VisualToolBench

Agentic Coding: GPT-5.4 Leads Public, Claude Leads Private

Structured comparison across SWE Atlas and SWE-Bench Pro subtasks.

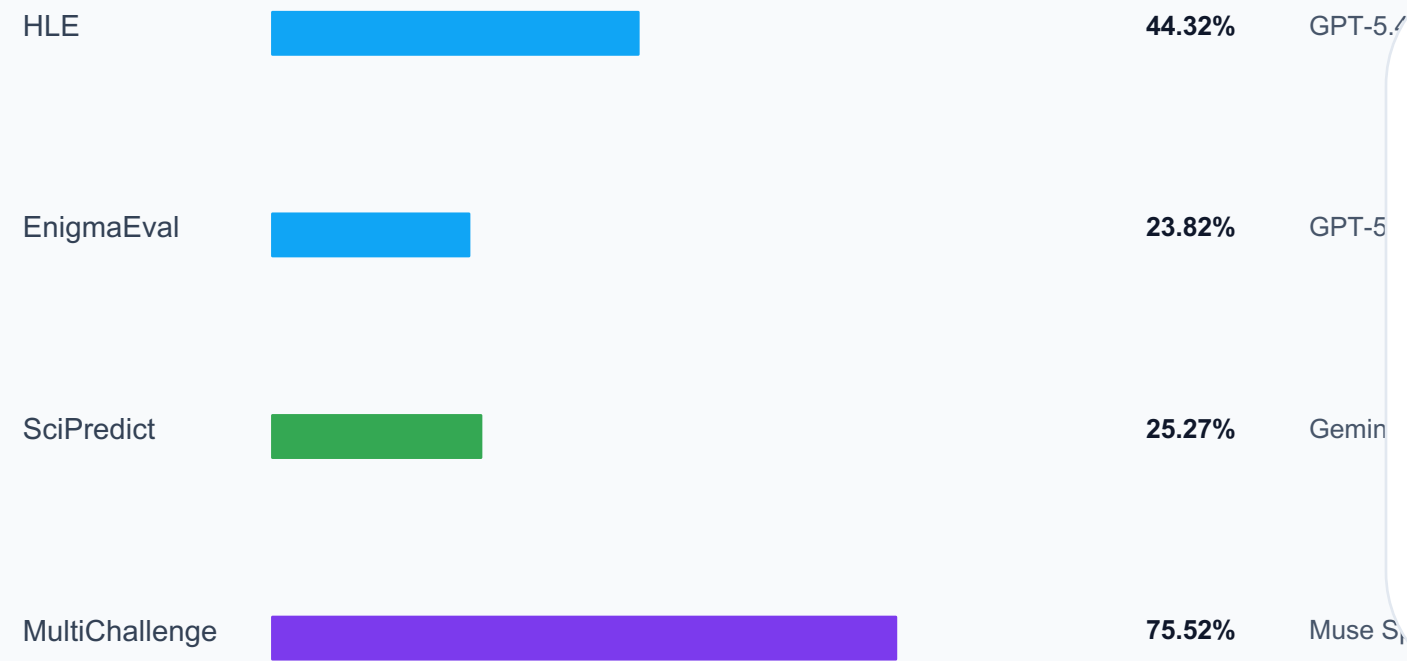
Benchmark	Rank 1	Score	Rank 2	Score	Rank 3	Score
SWE Atlas – Test Writing	GPT-5.4-xHigh (Codex CLI)	44.36	GPT-5.4-xHigh (Mini-SWE)	40.00	GPT-5.3-Codex-Xhigh	38.98
SWE Atlas – Codebase QnA	GPT 5.4 xHigh (Codex)	40.80	GPT 5.4 xHigh (Mini-SWE)	36.30	Opus 4.6 (Claude Code)	33.30
SWE-Bench Pro (Public)	GPT-5.4-pro (xHigh)	59.10	Muse Spark	55.00	Claude Opus 4.6	51.90
SWE-Bench Pro (Private)	Claude Opus 4.6	47.10	Muse Spark	44.70	GPT-5.4-pro (xHigh)	43.40

Takeaway: OpenAI dominates coding benchmarks overall, but Anthropic leads private-repo SWE-Bench Pro.

Frontier Reasoning: Progress with Large Unsaturated Gaps

Top benchmark scores remain far below ceiling on the hardest tasks.

Best Score by Benchmark (%)



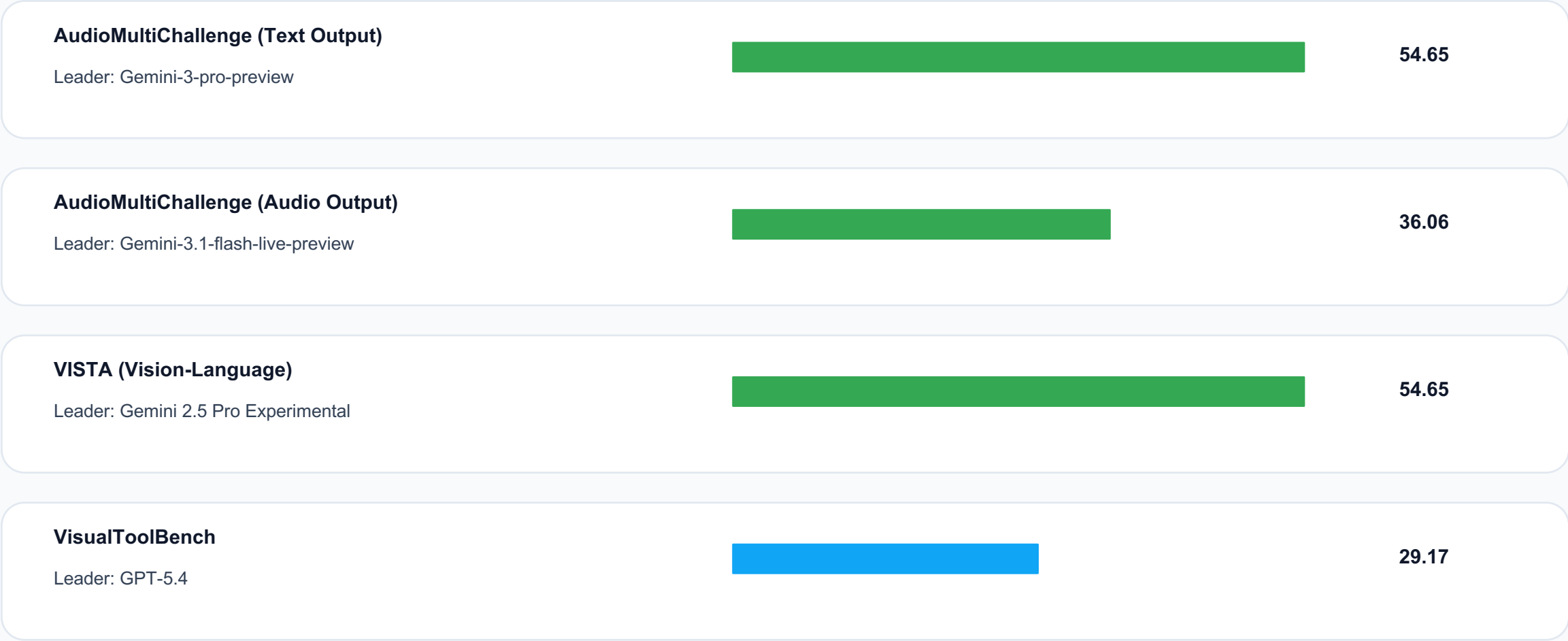
Interpretation

- HLE peak: 44.32
- EnigmaEval peak: 23.82
- SciPredict peak: 25.27
- MultiChallenge is higher (75.52) but still not near full reliability

Signal: frontier reasoning remains a major open problem.

Multimodal (Audio + Vision): Gemini Sweep with Narrow Visual Margin

Google leads most audio/vision boards; OpenAI remains competitive on VisualToolBench/VISTA.



Gemini holds all top-3 spots in AudioMultiChallenge (Text Output) and leads VISTA; GPT-5.4 narrowly leads VisualToolBench (29.17 vs 28.97).

Safety & Alignment: Near-Ceiling Honesty, Low Security Robustness

Contrasting MASK vs Fortress highlights large intra-domain variance.

MASK (higher is better)

Claude Opus 4.6: 96.28

Professional Reasoning

Legal: Claude Opus 4.6 at 52.27
Finance: Claude Opus 4.6 at 53.28

Fortress (higher is better)



Top score shown: 8.24

Claude Sonnet 4 shown at 12.80 in source table.

Interpretation: Alignment is not one-dimensional — honesty is near-saturated while national-security robustness remains weak.

Competitive Landscape: #1 Lab/Model by Benchmark

Heatmap-style matrix shows specialization across domains.

Benchmark	#1 Lab	#1 Model
SWE Atlas – Test Writing	OpenAI	GPT-5.4-xHigh (Codex CLI)
SWE Atlas – Codebase QnA	OpenAI	GPT 5.4 xHigh (Codex)
MCP Atlas	Muse Spark	Muse Spark
SWE-Bench Pro (Public)	OpenAI	GPT-5.4-pro (xHigh)
SWE-Bench Pro (Private)	Anthropic	Claude Opus 4.6
Humanity's Last Exam	OpenAI	GPT-5.4-pro
EnigmaEval	OpenAI	GPT-5.4-pro
MultiChallenge	Muse Spark	Muse Spark
SciPredict	Google	Gemini-3-pro-preview
AudioMultiChallenge (Text)	Google	Gemini-3-pro-preview
AudioMultiChallenge (Audio)	Google	Gemini-3.1-flash-live-preview
VISTA	Google	Gemini 2.5 Pro Experimental

Pattern: wins are distributed by domain; no lab is uniformly dominant.

Conclusion: Specialization Defines the Frontier Era

Decision implication for model selection, org strategy, and deployment architecture.

- OpenAI leads agentic coding and several frontier reasoning benchmarks (e.g., HLE 44.32, EnigmaEval 23.82).
- Google Gemini dominates multimodal audio/vision leaderboards (AudioMultiChallenge, VISTA).
- Anthropic Claude leads safety alignment and professional reasoning, with MASK at 96.28.
- Muse Spark is a credible new entrant with #1 on MultiChallenge (75.52) and MCP Atlas (78.30).
- No single model is best across all dimensions; benchmark-conditioned model routing is increasingly rational.

OpenAI

Google

Anthropic

Muse Spark